

Squareclass growth in structured rational sumsets

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Abstract

We give two consequences of established theorems that restrict certain approaches to large rational square-sum grids. First, among translates $1 + u$ of rational S -units, a fixed nonzero squareclass occurs at most $2^{16|S|+16}$ times. This follows from the Beukers–Schlickewei unit-equation bound. Second, a rational set of size M contained densely in a proper generalized arithmetic progression of fixed dimension d has at least $M^{2/(5d)}$ squareclasses, up to a logarithmic factor, in every translate. This follows by slicing into arithmetic progressions and applying Bombieri–Zannier. The Green–Ruzsa form of Freiman’s theorem gives a corresponding obstruction for sets with bounded additive doubling. We state all hypotheses and prove the deductions. Their originality has not been established. They do not resolve Erdős Problem 885.

1 Definitions and scope

For $t \in \mathbb{Q}^* = \mathbb{Q} \setminus \{0\}$, let $[t]$ denote its image in the group $\mathbb{Q}^*/\mathbb{Q}^{*2}$, viewed as a vector space over $\mathbb{F}_2$. Thus $[xy] = [x] + [y]$, and $[x] = [y]$ exactly when x/y is a nonzero rational square. Signs are included: a negative rational is not a square in $\mathbb{Q}$.

For a finite rational set T , define

$$T^* = T \setminus \{0\}, \quad c(T) = |\{[t] : t \in T^*\}|, \quad h(T) = \dim_{\mathbb{F}_2} \langle [t] : t \in T^* \rangle.$$

We set $c(\emptyset) = h(\emptyset) = 0$. For every such set,

$$c(T) \leq 2^{h(T)}. \tag{1}$$

Here $c(T)$ counts the classes present, whereas $h(T)$ measures the dimension of their span. Multiplication by a nonzero rational preserves $c(T)$; it need not preserve $h(T)$. In fact $|h(\lambda T) - h(T)| \leq 1$, since $[\lambda t] = [\lambda] + [t]$.

For finite sets $A, B \subset \mathbb{Q}$, write $A + B = \{a + b : a \in A, b \in B\}$. The associated grid has one row for each distinct element of A and one column for each distinct element of B . We are interested in families with $|A| = |B| = M \rightarrow \infty$, all $a + b \neq 0$, and

$$h(A + B) = o(\log M). \tag{2}$$

Section 4 explains why such a family would give arbitrarily large rational square-sum grids and a positive answer to the all-size assertion in Erdős Problem 885. The present note supplies no such family and no general obstruction for arbitrary rational sets.

All logarithms denoted $\log$ are natural; $\log_2$ denotes base two. Constants $C_B \geq 1$ and $\beta > 0$ below are absolute constants furnished by the Bombieri–Zannier theorem, enlarged once if necessary to cover short progressions.

2 Shifted rational units

A rational S -unit, for a finite set S of rational primes, is a number

$$u = \pm \prod_{p \in S} p^{e_p}, \quad e_p \in \mathbb{Z}.$$

No restriction is imposed on these exponents or on the sizes of the primes.

Theorem 2.1. *Let $2 \in S$, let $s = |S|$, and let U be a nonempty finite set of rational S -units with $-1 \notin U$. For every $q \in \mathbb{Q}^*$,*

$$|\{u \in U : 1 + u \in q\mathbb{Q}^{*2}\}| \leq 2^{16s+16}. \quad (3)$$

Consequently,

$$c(1 + U) \geq |U| 2^{-16s-16}, \quad h(1 + U) \geq \log_2 |U| - 16s - 16. \quad (4)$$

The bound is independent of q , including its prime support and height.

Proof. Fix q . For each solution $1 + u = qv^2$, where $v \in \mathbb{Q}^*$, set

$$K = \mathbb{Q}(\sqrt{q}), \quad \alpha = v\sqrt{q}, \quad x = \frac{1 + \alpha}{2}, \quad y = \frac{1 - \alpha}{2}.$$

Then $x + y = 1$ and $xy = -u/4 \neq 0$. Let S_K be the set of finite primes of K above the rational primes in S , and let E be the group of elements of K^* that are units at every finite prime outside S_K .

At any such prime, u is a unit, so $1 + u$ is integral. The equality $\alpha^2 = 1 + u$ implies that α is integral there. Since $2 \in S$, the numbers x, y are integral there as well. Their product $-u/4$ is a unit, so each is a unit. Thus $x, y \in E$. This argument does not add primes dividing q to S .

Suppose first that q is not a rational square. The nontrivial automorphism σ of the quadratic field K sends α to $-\alpha$. Therefore

$$(x, y) \in G = \{(\varepsilon, \sigma(\varepsilon)) : \varepsilon \in E\} \subset (\mathbb{C}^*)^2.$$

The map $E \rightarrow G$ is injective. By the classical S -unit rank formula, recalled in [4, Section 1.1],

$$\text{rank } G = \text{rank } E = r_1(K) + r_2(K) - 1 + |S_K| \leq 1 + 2s.$$

Here the ordinary unit rank of a quadratic field is at most one, and at most two finite primes lie above each rational prime.

Beukers–Schlickewei [1, Theorem 1.1] bounds the number of solutions of $x + y = 1$ in a rank- R finitely generated subgroup of $(\mathbb{C}^*)^2$ by 2^{8R+8} . Their theorem in fact holds in its divisible closure. Applying it to G gives at most 2^{16s+16} ordered pairs. Distinct values of u give distinct pairs, since $u = -4xy$. Choosing one of the two signs of v for each u suffices.

If q is a rational square, then x, y are rational S -units. The group of rational S -units has rank s , so its product with itself has rank $2s$. The same theorem gives the smaller bound 2^{16s+8} . This proves (3). Counting U by its classes and using (1) proves (4). $\square$

If -1 is allowed in U , remove it before applying the theorem: its translate is zero, which has no multiplicative squareclass. The logarithmic inequality then uses $\log_2 |U \setminus \{-1\}|$ and is asserted only when this set is nonempty.

2.1 An affine normalization for grids

Corollary 2.2. *Let $A, B \subset \mathbb{Q}$ be finite, with $|A| \geq 2$, $B \neq \emptyset$, and $a+b \neq 0$ for every $(a, b) \in A \times B$. Fix any $a_0 \in A$ and $b_0 \in B$, and define*

$$U_{a_0, b_0} = \left\{ \frac{a - a_0}{a_0 + b_0} : a \in A \setminus \{a_0\} \right\}. \quad (5)$$

Let S contain 2 and all primes appearing in the numerator or denominator of an element of U_{a_0, b_0} in lowest terms. Then

$$h(A + B) + 16|S| + 16 \geq \log_2(|A| - 1). \quad (6)$$

The corresponding statement with A and B interchanged also holds.

Proof. The set U_{a_0, b_0} has $|A| - 1$ distinct nonzero elements. An element equals -1 only if $a + b_0 = 0$, which is excluded. For its element indexed by a ,

$$1 + u = \frac{a + b_0}{a_0 + b_0}, \quad [1 + u] = [a + b_0] + [a_0 + b_0].$$

Both classes on the right belong to the span of the grid classes. Thus $h(1 + U_{a_0, b_0}) \leq h(A + B)$, and Theorem 2.1 proves the claim. $\square$

The normalized set in (5) is unchanged under

$$A \mapsto \lambda A + t, \quad B \mapsto \lambda B - t, \quad \lambda \in \mathbb{Q}^*, \quad t \in \mathbb{Q},$$

with the anchors transformed accordingly: both numerator and denominator acquire the factor λ . This invariance concerns U and its prime support; the unnormalized grid span dimension may change under scaling.

In a family satisfying (2), the smallest permissible S in every row normalization must satisfy

$$|S| \geq \left(\frac{1}{16} - o(1) \right) \log_2 M.$$

The same is true for every column normalization. Thus sublogarithmic normalized prime support cannot provide such a family. Prime support is not the same as the abstract multiplicative rank of a group: one rational generator can involve arbitrarily many primes. No bound with that smaller rank substituted for $|S|$ is asserted here.

3 Sets contained in proper progressions

We first record the arithmetic-progression consequence needed for slicing.

Lemma 3.1. *Let $t_n = a + nv$ for $0 \leq n < L$, with $a, v \in \mathbb{Q}$, $v \neq 0$, and integer $L \geq 1$. For every $q \in \mathbb{Q}^*$,*

$$|\{n : t_n \in q\mathbb{Q}^{*2}\}| \leq C_B L^{3/5} (\log(2L))^\beta. \quad (7)$$

The constants are independent of a, v, q .

Proof. Bombieri–Zannier [2, Theorem 1] gives this bound, with $\log L$ for $L \geq 2$, for square entries of a nonconstant integer arithmetic progression. Divide the present progression by q and choose a positive integer D clearing the denominators of a/q and v/q . Multiplication by D^2 gives an integer arithmetic progression. Every counted entry becomes a nonzero rational square that is an integer, hence an integer square. If necessary, reverse the progression and retain only its nonnegative terms; these form a contiguous progression of length at most L . The theorem applies to this progression, with the constant enlarged for lengths zero or one. This also covers negative q and either sign of v . $\square$

A rational generalized arithmetic progression is a set

$$P = \left\{ p_0 + \sum_{i=1}^d n_i v_i : 0 \leq n_i < L_i \right\}, \quad p_0, v_i \in \mathbb{Q}, \quad L_i \in \mathbb{Z}_{\geq 2}. \quad (8)$$

It is *proper* if all the displayed representations are distinct. In that case its volume $V = \prod_i L_i$ equals $|P|$. Length-one directions can always be removed.

Theorem 3.2. *Let $A \subset \mathbb{Q}$ have size $M \geq 2$. Suppose $A \subset P$, where P is the proper progression (8), $d \geq 1$, and $V \leq C_0 M$ for some $C_0 \geq 1$. Put $L = \max_i L_i$. For any $b \in \mathbb{Q}$, let $z = 1$ if $0 \in A + b$ and $z = 0$ otherwise. Then*

$$c(A + b) \geq \frac{(M - z)L^{2/5}}{C_B V (\log(2L))^\beta}. \quad (9)$$

In particular, define

$$\kappa(C_0) = \frac{1}{2C_B C_0 (1 + \log C_0 / \log 4)^\beta} > 0.$$

Then

$$c(A + b) \geq \frac{\kappa(C_0) M^{2/(5d)}}{(\log(2M))^\beta}, \quad (10)$$

$$h(A + b) \geq \frac{2}{5d} \log_2 M - \beta \log_2 \log(2M) + \log_2 \kappa(C_0). \quad (11)$$

These bounds permit a zero in the translate and are uniform in all rational steps, offsets, denominators, and shifts.

Proof. Fix a direction of length L , and fix the other $d - 1$ coordinates. Properness partitions $P + b$ into exactly V/L disjoint arithmetic progressions of length L . Their common step is nonzero, since $L \geq 2$ and P is proper. By Lemma 3.1, any one class occurs at most

$$\frac{V}{L} C_B L^{3/5} (\log(2L))^\beta = C_B V L^{-2/5} (\log(2L))^\beta$$

times in $P + b$, hence at most this many times in its subset $A + b$. There are $M - z$ nonzero elements to count. Division gives (9). No density assumption on any individual slice of A is needed.

Since $M \leq V \leq C_0 M$,

$$L \geq V^{1/d} \geq M^{1/d}, \quad L \leq C_0 M, \quad M - z \geq M/2.$$

Moreover, for $M \geq 2$,

$$\log(2L) \leq \log(2M) + \log C_0 \leq \left(1 + \frac{\log C_0}{\log 4}\right) \log(2M).$$

Substitution proves (10). Taking base-two logarithms and using (1) proves (11). $\square$

3.1 Bounded additive doubling

Corollary 3.3. *For each fixed $K \geq 1$ there are an integer $D(K) \geq 1$ and a constant $C(K)$ such that, whenever $|A| = M \geq 2$ and $|A + A| \leq KM$, every $b \in \mathbb{Q}$ satisfies*

$$h(A + b) \geq \frac{2}{5D(K)} \log_2 M - \beta \log_2 \log(2M) - C(K). \quad (12)$$

Consequently $h(A + b) \geq c_K \log M$ for all sufficiently large M , with $c_K > 0$ depending only on K .

Proof. The Green–Ruzsa theorem [3, Theorem 1.1] puts A inside an actual proper coset progression in its ambient group, of dimension at most $D(K)$ and size at most $F(K)M$. Its stated estimates allow

$$D(K) = \max\{1, \lceil CK^4 \log(K + 2) \rceil\}, \quad F(K) = \exp(CK^4 \log^2(K + 2))$$

for a sufficiently large absolute C . The additive group $\mathbb{Q}$ has no nontrivial finite subgroup, so the finite subgroup part of the coset progression is trivial. We therefore obtain a proper rational progression as in Theorem 3.2, with $d \leq D(K)$ and $C_0 = F(K)$. Its actual rational values are retained; no change by an arbitrary Freiman isomorphism is being applied to squares. Equation (11) proves the result, taking $C(K) = -\log_2 \kappa(F(K))$. $\square$

If a balanced family satisfies (2), both

$$\frac{|A + A|}{|A|} \rightarrow \infty, \quad \frac{|B + B|}{|B|} \rightarrow \infty. \quad (13)$$

Indeed, if the first ratio failed to tend to infinity, some infinite subsequence would have a fixed upper bound K . Choose any one element $b \in B$ for each grid on that subsequence. The span of its column classes is contained in the full grid span, so Corollary 3.3 contradicts (2). Interchanging A, B proves the other assertion.

4 Connection with Erdős Problem 885

For a positive integer N , put

$$D(N) = \{|r - s| : r, s \text{ are positive integers and } rs = N\}.$$

The all-size question asks whether, for every positive integer k , there are distinct positive integers $N_1, \dots, N_k$ for which $\bigcap_{i=1}^k D(N_i)$ contains at least k distinct numbers [5]. We include the elementary connections to identify exactly the scope of the preceding obstructions.

Proposition 4.1. *For a fixed k , the preceding assertion is equivalent to the existence of rational sets A, B of sizes $k + 1, k$, respectively, with all cross sums rational squares, allowing zero. Thus its truth for every k is equivalent to the existence of arbitrarily large balanced rational square-sum grids whose entries are nonzero.*

Proof. Given the integers and k distinct common differences $d_j \geq 0$, take

$$A = \{0, 4N_1, \dots, 4N_k\}, \quad B = \{d_1^2, \dots, d_k^2\}.$$

For any factor pair $rs = N_i$ with $|r - s| = d_j$, one has $d_j^2 + 4N_i = (r + s)^2$. Every cross entry, including those in row zero, is a square.

Conversely, order the rows $a_0 < a_1 < \dots < a_k$. For each b_j , choose nonnegative rational roots r_j, s_{ij} with

$$r_j^2 = a_0 + b_j, \quad s_{ij}^2 = a_i + b_j.$$

Let the positive integer L clear all root denominators, and set

$$N_i = L^2(a_i - a_0), \quad d_j = 2Lr_j.$$

The numbers N_i are distinct positive integers, because each is the difference of the integer squares $(Ls_{ij})^2$ and $(Lr_j)^2$. Their positive integer factorizations

$$N_i = (L(s_{ij} - r_j))(L(s_{ij} + r_j))$$

have differences d_j . These differences are distinct, since the distinct b_j give distinct nonnegative r_j .

An arbitrarily large balanced square grid contains every required $(k+1)$ -by- k rectangle. Conversely the all-size assertion gives arbitrarily large balanced grids by restriction. To remove zeros, start with a $2m$ -by- $2m$ square grid, select any m rows, and discard the columns having a zero in these rows. Each row has at most one such column, so at least m columns remain. This gives a zero-free m -by- m grid. $\square$

Lemma 4.2. *Let A, B each have M distinct rational elements, with all cross sums nonzero. Suppose at most $r \geq 1$ squareclasses occur. If*

$$M \geq \max\{2kr, k(2r)^k\}, \tag{14}$$

then some k -by- k subrectangle has a single squareclass. Dividing both coordinate sets of this subrectangle by a representative of that class makes every entry a nonzero rational square.

Proof. Color each of the M^2 positions by its squareclass. Some color occurs at least M^2/r times. Let d_b be its degree at column b . Discrete convexity of $n \mapsto \binom{n}{k}$ on the nonnegative integers, with $\binom{n}{k} = 0$ for $n < k$, gives

$$\sum_{b \in B} \binom{d_b}{k} \geq M \binom{\lfloor M/r \rfloor}{k}.$$

Hence some k -element row set has at least

$$\frac{M \binom{\lfloor M/r \rfloor}{k}}{\binom{M}{k}} \geq \frac{M}{(2r)^k} \geq k$$

common columns of this color. For the middle inequality, every factor in the binomial ratio is at least $1/(2r)$ because $M/r \geq 2k$. Selecting k common columns proves the claim. If their class is $[q]$, each entry has the form qv^2 , so $(a/q) + (b/q) = v^2$. This works for negative q as well. $\square$

Under (2), equation (1) gives $r \leq 2^{h(A+B)} = M^{o(1)}$. For each fixed k , condition (14) eventually holds. For the original assertion at size k , apply Lemma 4.2 at size $k+1$ and retain k of its columns. Proposition 4.1 then gives the desired integers.

5 What remains unresolved

The preceding deductions require substantial structure. Theorem 2.1 requires control of the prime support of a specific normalized coordinate set. Theorem 3.2 requires containment with bounded relative volume in a proper progression of fixed dimension. Corollary 3.3 supplies this containment when additive doubling is bounded.

Nothing here shows that arbitrary rational grids meet either structural condition. A possible family satisfying (2) must have both the normalized prime-support growth from Corollary 2.2 and the unbounded doubling in (13); these are necessary conditions, not a contradiction. Conversely, this note constructs no family in that remaining regime. In particular, the unrestricted question in Erdős Problem 885 remains unresolved by this work.

Attribution and review status. The inputs are published results of Beukers and Schlickewei, Bombieri and Zannier, and Green and Ruzsa, together with the classical S -unit rank formula. The propositions here are presented as deductions from those results; no claim of originality or priority is made. AI assistance was used for literature retrieval, proof development, drafting, and consistency checks. This draft has not undergone peer review. Small exact-arithmetic checks of normalization identities, if supplied alongside it, are consistency tests rather than proofs of the cited theorems or of the unrestricted problem.

References

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